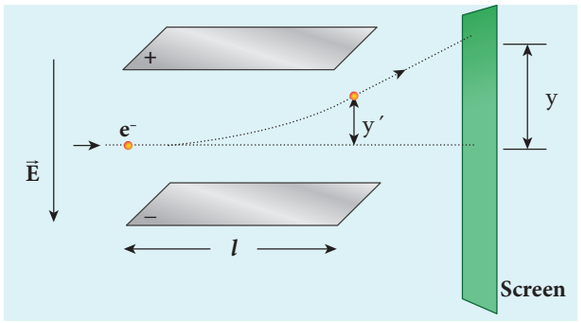


Figure 9.5 Deviation of path by applying uniform electric field

Let y be the deviation produced from original position on the screen as shown in Figure 9.5. Let the initial upward velocity of cathode ray be $u = 0$ before entering the parallel electric plates. Let t be the time taken by the cathode rays to travel in electric field. Let l be the length of one of the plates, then the time taken is

$$t = \frac{l}{\upsilon} \qquad (9.5)$$

Hence, the deflection y' of cathode rays is (note: $u = 0$ and $a_e = \frac{e}{m}E$)

$$y' = ut + \frac{1}{2}at^2 \Rightarrow y' = ut + \frac{1}{2}a_e t^2$$
$$= \frac{1}{2}\left(\frac{e}{m}E\right)\left(\frac{l}{\upsilon}\right)^2$$

$$y' = \frac{1}{2}\frac{e}{m}\frac{l^2B^2}{E} \qquad (9.6)$$

Therefore, the deflection y on the screen is

$$y \propto y' \Rightarrow y = Cy'$$

where C is proportionality constant which depends on the geometry of the discharge tube and substituting y' value in equation 9.6, we get

$$y = C\frac{1}{2}\frac{e}{m}\frac{l^2B^2}{E} \qquad (9.7)$$

Rearranging equation (9.7) as

$$\frac{e}{m} = \frac{2yE}{Cl^2B^2} \qquad (9.8)$$

Substituting the values on RHS, the value of specific charge is calculated as $\frac{e}{m} = 1.7 \times 10^{11}\,\mathrm{C\,kg^{-1}}$.

Note

The specific charge is independent of
(a) gas used
(b) nature of the electrodes

9.2.2 Determination of charge of an electron – Millikan's oil drop experiment

S0382

Millikan's oil drop experiment is another important experiment in modern physics which is used to determine one of the fundamental constants of nature known as charge of an electron (Figure 9.6 (a)).

By adjusting electric field suitably, the motion of oil drop inside the chamber can be controlled – that is, it can be made to move up or down or even kept balanced in the field of view for sufficiently long time.

The experimental arrangement is shown in Figure 9.6 (b). The apparatus consists of two horizontal circular metal plates A and B each with diameter around 20 cm and are separated by a small distance 1.5 cm. These two parallel

XII_Physics_Vol-2_UNIT-9_EM.indd 145 19-12-2022 16:04:35